Journal of Machine Learning Research 26 (2026) 1-6 Submitted 4/26; Revised 5/26; Published 9/26

A Causal Evaluation of Timeouts in the NBA

Aksel Kretsinger-Walters ADK2164@COLUMBIA.EDU
Department of Computer Science
Columbia University
New York, NY 10027, USA

Editor: David M. Blei

Abstract

There is a long-standing belief in the NBA that calling a timeout during an opponent's scoring run can disrupt their momentum and improve the calling team's chances of winning. However, the empirical evidence in support of this belief is limited and inconclusive. In this paper, we use play-by-play data from the 2021-2026 NBA seasons to conduct a causal analysis of the effect of timeouts on team performance.

Keywords: Causal inference, sports analytics, NBA, timeouts, momentum, difference-in-differences, synthetic control

1 Related Works

The question of whether timeouts actually disrupt momentum has been studied from several angles, with notably conflicting conclusions. Assis et al. (2020) apply a formal causal framework to NBA play-by-play data, using matching techniques informed by a causal graph to estimate the counterfactual effect of a timeout. They conclude that timeouts have *no* effect on team performance, and argue that the apparent rebound observed after a timeout is simply regression to the mean: because timeouts tend to be called when the opposing team is scoring at an unusually high rate, any subsequent slowdown reflects the natural tendency of scoring to return to its average rather than a causal effect of the stoppage itself.

Gibbs et al. (2021) revisit the same question but restrict attention to the specific scenario of an opposing team's scoring run, where the strategic stakes of the timeout decision are highest. Working within the Rubin causal model, they carefully define units under SUTVA and introduce an interpretable outcome based on the score difference to capture broad changes in scoring dynamics. In contrast to Assis et al. (2020), they find that calling a timeout during an opposing run is *slightly disadvantageous* on average, and further show that the magnitude of this effect varies meaningfully by franchise. Their result suggests that comebacks after runs happen frequently enough to create the illusion of timeout efficacy, but that the intervention itself does not help.

Weimer et al. (2023) take a different identification strategy by exploiting mandatory television timeouts, which are triggered at predetermined points in the game and are therefore plausibly exogenous with respect to in-game momentum. Using TV timeouts as a natural experiment that mimics random assignment of a pause, they report an *11.2% decline* in the scoring rate of the team carrying momentum immediately after a TV timeout, and they show this effect is robust to run size, substitutions, and game context. This stands in sharp contrast to the null result of Assis et al. (2020) and the mildly negative result of Gibbs et al. (2021): by removing coach selection bias entirely, Weimer et al. (2023) argue that team-level momentum genuinely exists and that pauses in play do interrupt it.

Qiu et al. (2025) approach the problem from a sports-psychology perspective rather than a causal-inference one, analyzing 4,051 timeouts across 1,235 elite basketball games using linear mixed models to test pre- versus post-timeout momentum over short (48s), medium (96s), and long (144s) windows. They find that timeouts significantly *increase* momentum for teams that are trailing or otherwise disadvantaged, while tending to *decrease* momentum for teams that are already ahead, with contextual factors such as game period and opponent strength substantially moderating the effect. Unlike the three causal papers above, which target an average treatment effect, their results emphasize heterogeneity: timeouts are neither uniformly helpful nor uniformly useless, and their efficacy is highly conditional on game state. Together these four studies motivate our own analysis, which aims to reconcile the tension between the exogenous-timeout evidence for a real momentum effect and the selection-bias-corrected evidence against one.

2 Methods

2.1 Data

We build our analysis on the cdnnba enriched play-by-play data for the 2020–2025 NBA seasons (regular season and playoffs), loaded via our own `CDNNBAMemoPL.load_all()` pipeline. After cleaning, the spine contains 4,183,347 play-by-play rows spanning 7,457 unique games and 1,480,697 possessions, of which 83,662 are coach-called timeouts and 14,561 are inferred mandatory television timeouts.

2.2 Timeout classification

We partition stoppage events into three mutually exclusive groups. **Endogenous** timeouts are coach-initiated stoppages, identified by `actionType == timeout` and `subType` in {`full`, `challenge`}. These are endogenous in the causal sense because coaches decide when to call them in response to game state. **Exogenous** timeouts are the mandatory television timeouts specified by the NBA rulebook at roughly the 6:59 and 2:59 marks of each period. The cdnnba feed does not directly label TV timeouts, so we inject them using a rulebook-based inference procedure and assign them `subType == official_inferred`. We validated this procedure against the explicitly labeled 2013–2016 nbastatsv3 feed and obtained an F1 of 0.88–0.92, indicating good but imperfect recovery of mandatory timeouts. **Control** events are dead-ball moments with no timeout occurring during an active scoring run; to avoid over-representing long runs, we subsample at most one control per run segment.

2.3 Run and recovery definitions

A *scoring run* at play t is defined as $|\texttt{streak}(t)| \geq s$, where $\texttt{streak}(t)$ is the cumulative net score swing within the current scoring segment, computed by resetting the counter whenever the scoring team flips. We refer to the team being outscored as the *suffering team* and the team scoring as the *running team*.

Our primary outcome is the *recovery* metric: the signed lead change from the suffering team's perspective over a forward window of n minutes. If $\texttt{lead}(t)$ is the home-minus-away margin at play t, we define

$$\texttt{recovery}(t, n) = -\operatorname{sign}(\texttt{streak}(t)) \cdot \big(\texttt{lead}(t + n \text{ min}) - \texttt{lead}(t)\big),$$

so that positive values mean the suffering team clawed points back during the window. As a secondary outcome we also report `running_team_pts`, the raw points scored by the running team over the forward window, which matches the dependent variable used by Weimer et al. (2023).

2.4 Between-group estimators

As a first pass we compute group means of `recovery` conditional on run size s and forward window n, and report differences versus the control group using Welch's t-tests. We sweep $s \in \{4, 5, 6, 7, 8, 10, 12, 15\}$ and $n \in \{0.5, 1, 1.5, 2, 3, 4, 5\}$ minutes, and also stratify by quarter, regular season vs. playoffs, and absolute score margin.

2.5 Within-game matched-twin causal design

The between-group estimator is vulnerable to game-level confounding: coaches call timeouts in systematically different games than the ones where control events are drawn, and mandatory TV timeouts likewise cluster in particular game types. To remove this bias we use a *within-game matched-twin* design. For each treated event (endogenous or exogenous timeout during a run of size $\geq s$), we search for a control event *in the same game* that also satisfies: the same period; the same sign of `streak` (so the run is against the same team); $|\Delta\,\texttt{streak}| \leq 2$ (comparable run magnitude); $|\Delta\,\texttt{seconds_remaining}| \leq 120$s (comparable clock position within period); and $|\Delta\,\texttt{suffering\_margin}| \leq 3$ (comparable score state). When multiple candidate controls match, we pick the one minimizing a combined distance over those dimensions. The per-pair causal estimate is the difference in `recovery` between the treated event and its matched twin, and we aggregate across pairs with a paired t-test. This design holds team composition, pace, coaching, and game-level momentum approximately fixed by construction.

2.6 Moderator analyses

To probe heterogeneity we repeat the between-group and matched-twin analyses within slices defined by (i) timeout subtype (`coach_full`, `coach_challenge`, `official_inferred`, `stoppage`); (ii) game phase and clutch status (last five minutes of Q4 with margin ≤ 5); (iii) season-long team quality gap, using each team's `NET_RATING` from `player_advanced_stats`; (iv) the number of substitutions within a ± 30s game-clock window around the event, bucketed as $\{0,\ 1\text{–}2,\ 3+\}$; and (v) the running team's in-game cumulative points-per-possession (PPP) at the time of the event, as a proxy for short-term shooting rhythm.

3 Evaluation

3.1 Between-group results

The simple between-group comparison essentially reproduces the weak-to-null effects reported in the prior literature. Across run sizes $s \in \{5, \ldots, 10\}$ and forward windows $n \in \{1, \ldots, 5\}$ minutes, the endogenous-vs-control gap in `recovery` is never statistically significant (all $|\Delta| < 0.1$ pts, $p > 0.05$ in 15 of 16 cells). Exogenous timeouts, by contrast, look significantly *negative*, with $\Delta \approx -0.25$ to -0.45 pts at the 3-minute window depending on run size ($p < 0.001$). Sweeping the forward window at fixed run size $s = 6$ shows the exogenous penalty growing from -0.20 at 30 seconds to a plateau around -0.27 at 3 minutes, while the endogenous gap hovers near zero throughout. At face value this matches the "timeouts don't help (and mandatory TV breaks hurt)" consensus. As a secondary check we also computed `running_team_pts` in the style of Weimer et al. (2023) and found a *positive* exogenous-minus-control gap of +2.8% to +14.6% depending on window — the opposite sign from Weimer's reported −11.2%. This is internally consistent with our suffering-team analysis (if the suffering team recovers less, the running team must score relatively more) and likely reflects the post-2017 mandatory-timeout rule changes together with the inferred-rather-than-labeled nature of our TV timeouts.

3.2 Within-game matched-twin results

The matched-twin design, which is our cleanest causal test, reverses the headline conclusion of the between-group analysis. With $s = 6$ and a 3-minute window we obtain 8,906 matched endogenous pairs and 2,667 matched exogenous pairs. The paired mean differences are +0.397 **pts for endogenous timeouts** ($t = 27.9$, $p < 0.0001$) and +0.455 **pts for exogenous timeouts** ($t = 11.8$, $p < 0.0001$). Both effects are highly significant and both favor the suffering team; the exogenous effect is in fact slightly *larger* than the endogenous effect, which is consistent with coaches selectively burning timeouts in especially adverse game states where recovery is hard even with help. Breaking the treated population into fine subtypes, every subtype shows the same sign and roughly the same magnitude: `coach_full` +0.396 ***, `coach_challenge` +0.714 * (small sample, $n = 35$), `official_inferred` +0.449 ***, and generic `stoppage` events +0.458 ***. The reversal relative to the between-group results is driven by game-level selection: coach timeouts are concentrated in low-recovery games, so comparing the endogenous mean to a global control mean absorbs the selection into the baseline and erases the signal. Matching within game cancels this bias by construction, and a simpler within-game mean comparison gives a consistent +0.35 pts effect for endogenous timeouts with the exogenous effect shrinking to near zero.

3.3 Mechanism and moderators

Once we have a signal, we can ask where it comes from. Stratifying by the number of substitutions in a ±30s window around the event reveals that the endogenous effect is concentrated entirely in timeouts accompanied by personnel changes: with 0 substitutions the endogenous-control gap is −0.05 (n.s.), with 1–2 substitutions it is +0.10 (n.s.), and with 3 or more substitutions it rises to +0.21 ($p < 0.001$). The exogenous penalty is correspondingly largest (-0.39, $p < 0.001$) when no subs occur and vanishes when many do. This directly confirms the mechanism proposed by Weimer et al. (2023): the value of a timeout is mediated by substitutions rather than by the pause itself. Conditioning on team quality via season-long `NET_RATING`, we find that the endogenous benefit is strongest in evenly matched games ($+0.23$ pts, $p < 0.05$) and washes out in lopsided matchups where the talent gap dominates the trajectory. Finally, stratifying by the running team's in-game PPP we find that when the opponent is shooting unusually efficiently (PPP ≥ 1.30), both effects are amplified and point in opposite directions: endogenous timeouts produce $+0.24$ pts of extra recovery ($p < 0.05$) while exogenous timeouts produce -0.62 pts ($p < 0.001$). This is the clearest evidence in our data of a genuine momentum effect — a coach timeout with substitutions disrupts a hot-shooting opponent, while a pure TV break effectively gives that opponent a rest and lets the run continue.

3.4 Synthesis

Our headline finding is that timeouts do causally improve the suffering team's short-term recovery — by roughly $+0.4$ points over the following three minutes — and that this effect is masked in between-group analyses by a subtle but severe game-level selection bias. The published consensus of "no effect" or "slightly negative effect" (Assis et al., 2020; Gibbs et al., 2021) reproduces in our data only when we use a between-group estimator; the within-game matched-twin estimator, which holds game identity approximately fixed by construction, produces a strongly positive and highly significant effect for both endogenous and exogenous stoppages. The benefit is overwhelmingly mediated by substitutions (in agreement with Weimer et al. (2023)) and is largest in evenly matched games and against hot-shooting opponents, which matches both coaching intuition and the conditional-momentum findings of Qiu et al. (2025). The pause alone does essentially nothing; the pause plus a lineup change is worth roughly a third of a possession of expected recovery.

Acknowledgments and Disclosure of Funding

N/A

References

Niander Assis, Renato Assunção, and Pedro O. S. Vaz-de Melo. Stop the clock: Are timeout effects real? In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*, 2020. URL https://arxiv.org/abs/2009.06750. arXiv:2009.06750.

Connor Gibbs, Ryan Elmore, and Bailey Fosdick. The causal effect of a timeout at stopping an opposing run in the NBA. *arXiv preprint arXiv:2011.11691*, 2021. URL https://arxiv.org/abs/2011.11691.

Mingjia Qiu, Kaimin Zhang, Yafei Chao, and Mingxin Zhang. Interrupt or reinforce? the impact of timeout on momentum in basketball game. *Frontiers in Psychology*, 16:1673186, 2025. doi: 10.3389/fpsyg.2025.1673186. URL https://doi.org/10.3389/fpsyg.2025.1673186.

Louis Weimer, Zachary C. Steinert-Threlkeld, and Kevin Coltin. A causal approach for detecting team-level momentum in NBA games. *Journal of Sports Analytics*, 9(2):117–132, 2023. doi: 10.3233/JSA-220592. URL https://doi.org/10.3233/JSA-220592.